

36118 Applied Natural Language Processing
Master of Data Science and Innovation, UTS

Assessment Task 2B

End-to-End NLP Project — Technical Report

Sydney Liveability Explorer

Subject	36118 Applied Natural Language Processing
Submission	AT2 Part B — Group Technical Report
Due Date	Monday, 4 May 2026, 23:59
Suggested Length	10–15 pages (excluding appendix)
Group	Group [Number], Autumn 2026
GitHub Repository	<a href="https://github.com/<org-or-user>/<repo>">https://github.com/<org-or-user>/<repo>

Team Members

Last Name	First Name	Student ID	GitHub Username
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LLM-Assisted Progress Log

This log is intended for incremental drafting with LLMs. Add one row per meaningful update so the report evolution is auditable and reproducible.

Date	Author	Section	Status	What Changed	Next Action
2026-04-23	LLM	Template Init	Done	Initialized LaTeX report structure from DOCX template.	Start drafting Section 1 with verified project outcomes.
2026-04-24	LLM	Executive Summary	Draft	Added full Executive Summary: problem statement, data sources, technical stack, and key outcomes.	Review with team and confirm project outcomes are accurate.

1 Executive Summary

The **Sydney Liveability Explorer** addresses a practical challenge faced by prospective residents, students, and urban planners: how to compare Sydney suburbs across multiple liveability dimensions in a unified, data-driven way. Existing tools present individual metrics in isolation, forcing users to manually cross-reference safety statistics, transport access, and community sentiment without a coherent synthesis.

This project delivers an end-to-end NLP pipeline and interactive web application that integrates six structured data sources — Reddit community posts, BOCSAR crime records, ArcGIS civic facilities, OpenStreetMap amenities, transport infrastructure data, and government PDF community reports — to produce suburb-level liveability scores and conversational answers.

The technical stack combines **traditional NLP** (VADER sentiment, LDA topic modelling, TF-IDF keyword extraction) as interpretable baselines with **modern approaches** (DistilRoBERTa aspect-level sentiment, sentence embeddings via all-MiniLM-L6-v2, and Retrieval-Augmented Generation via CrewAI and ChromaDB). A FastAPI backend exposes a weighted scoring API and a conversational chat endpoint, consumed by a Next.js frontend featuring an interactive Leaflet map, suburb comparison charts, and a preference-driven onboarding flow.

Key outcomes include a functional multi-agent query crew that routes user questions to specialist data retrievers and synthesises grounded answers, a weighted liveability ranking engine covering over 30 Sydney suburbs, and a reproducible ingestion pipeline for all six data sources.

2 Project Objectives and Scope

Define the primary objectives, intended user group, and boundaries of this submission.

3 Project Phases and Team Contributions

Describe project phases and map responsibilities to each team member.

Phase Breakdown Table

Phase	Tasks	Team Member(s)
Phase 1 — Data	Data acquisition and preprocessing from Reddit, PDF, BOCSAR, ArcGIS, OSM, and transport sources.	
Phase 2 — NLP Pipeline	Aspect-level sentiment, TF-IDF extraction, topic modeling, preprocessing pipeline.	
Phase 3 — RAG + LLM	Embeddings, ChromaDB vector store, CrewAI query crew, FastAPI backend.	
Phase 4 — Frontend	Next.js dashboard, Leaflet map, charts, conversational UI, onboarding weights.	
Phase 5 — Evaluation + Report	Traditional vs modern NLP comparison, metrics, report, poster, peer review.	

4 Data Sources

Describe each source, access method, format, granularity, and known limitations.

4.1 Primary Data Sources

Document each source contribution to the pipeline.

4.2 Data Limitations and Ethical Considerations

Discuss data bias, staleness, privacy, consent, and social media ethics.

5 NLP Methods and Techniques

Explain methods, rationale, implementation details, and reproducibility settings.

5.1 Text Preprocessing

Tokenization, stopword removal, lemmatization, and domain-specific cleaning.

5.2 Traditional NLP Approaches

5.2.1 Sentiment Analysis (VADER + TextBlob) — Baseline

Explain baseline polarity and subjectivity workflow.

5.2.2 Topic Modelling (LDA — Gensim)

Document topic count, parameters, labeling strategy, and coherence metrics.

5.2.3 TF-IDF Keyword Extraction

Describe extraction strategy and baseline value.

5.3 Modern NLP Approaches

5.3.1 Sentence Embeddings (all-MiniLM-L6-v2)

Explain chunking strategy and ChromaDB upsert details.

5.3.2 Retrieval-Augmented Generation (RAG + LLM)

Detail query crew orchestration, retrieval filters, and grounded answer generation.

5.3.3 Aspect-Level Sentiment Analysis (DistilRoBERTa)

Describe aspect taxonomy, emotion profiling, and production rationale.

5.4 Traditional vs. Modern NLP Comparison

Compare representative query outputs and discuss trade-offs.

6 Findings and Evaluation

Present core findings and evidence from notebooks and backend outputs.

6.1 NLP Pipeline Results

Summarize aspect scores, emotions, topics, and keyword patterns.

6.2 Model Evaluation

Provide metrics and qualitative error analysis.

6.3 Application Evaluation

Report usability observations from walkthroughs or testing.

6.4 Temporal Trends

Summarize relevant time-based patterns from crime and sentiment data.

7 Outcomes and Value Added

Describe user-facing outcomes and practical value for target groups.

8 Recommendations and Next Steps

List highest-impact extensions and maintenance priorities.

9 Conclusion

Summarize contributions, lessons learned, and future potential.

10 References

Use APA 7th edition for all citations.

A Appendix A — System Architecture Diagram

Insert architecture diagrams and explain key flows.

B Appendix B — Dashboard Screenshots

Insert annotated screenshots of map, charts, chat, onboarding, and debug views.

C Appendix C — Sample Code

Add representative code snippets for ingestion, embeddings, agents, and API endpoints.

D **Appendix D — Evaluation Tables and Charts**

Include confusion matrices, coherence tables, and charts used in Section 6.

E **Appendix E — Traditional vs. Modern NLP Comparison Table**

Table 3: Traditional vs. Modern NLP Comparison

Technique	Type	Speed	Interpretability	Role in This Project
TF-IDF Keyword Extraction	Traditional	Fast	High	Baseline retrieval — top terms per suburb
LDA Topic Modelling	Traditional	Medium	Medium	Thematic discovery baseline
VADER + TextBlob	Traditional	Fast	High	Baseline sentiment benchmark
RAG + LLM (CrewAI + ChromaDB)	Modern	Medium	Medium	Grounded, cited conversational answers
DistilRoBERTa Aspect Sentiment	Modern	Medium	Medium	Production sentiment and emotion profiling